Textual Networks of St. Wenceslas: The application of HTR and LLM tools to Church Slavonic Hagiographical Sources

Olga Kalashnikova*[1]

[1]Università degli studi di Napoli-L'Orientale (Orientale) – Italy

Abstract

Digital humanities methods have transformed medieval studies, but Church Slavonic (CS) textual traditions remain underrepresented in computational approaches due to limited digitized corpora. The paper demonstrates the limits of combining Handwritten Text Recognition (HTR), Large Language Models (LLMs), and network analysis in examining textual transmission of CS hagiographical sources in the case of the liturgical office for St. Wenceslas of Bohemia (d. 935) preserved in three 11th-12th-century manuscripts (referred here as Sin, Sof, and T).

In order to test a hypothesis that network visualization of the office's textual variants can reconstruct the source's manuscript genealogy (stemma codicum) and transmission intricacies, I followed Carson Koepke's approach to textual and manuscript networks and conducted three comparative tests. First, a bipartite network mapped offices for which saints appear in each manuscript's content. The second bipartite network compared the structural organization of the liturgical office for Wenceslas itself across the three witnesses. These networks, however, proved less informative because they revealed that all three manuscripts clustered tightly with nearly identical overall liturgical organization and structure of the office for Wenceslas (Figures 1 and 2).

Next, three CS textual witnesses of the office for Wenceslas were transcribed using a domain-specific HTR model, EastSlavCyrillic_11-12v2, on Transkribus. Claude-based entity recognition prompt was subsequently used to generate a collation CSV file documenting 594 textual variation units across all 41 verses of the office. The LLM extracted variant readings at each locus, which were then manually verified and classified by type (orthographic, substantive, lexical error etc.). Gephi visualization mapped pairwise agreement percentages between the textual witnesses as edge weights (Sin-Sof: 56.1%; T-Sof: 49.5%; Sin-T: 48.0%) and error counts as node attributes (Sin: 17; T: 11; Sof: 10). The resulting network positioned Sin and Sof as a genealogically related cluster with the highest textual agreement, while T remained isolated, thus representing a slightly independent textual tradition. Node sizing by error frequency identified Sin as the most corrupted witness and Sof as the most reliable despite sharing ancestry with Sin (Figure 3).

The obtained automated results corroborate the most recent philological conclusions about the text's bifurcated transmission. According to V.S. Tomelleri, Sin and Sof must have descended from a common archetype that contains shared errors (confirmed at loci V5.4, V19.3, V20.3 in the collation), while T preserves an independent recension often closer to

*Speaker

sciencesconf.org:hnr2026:746626

Greek sources. The HTR-mined and LLM-run network approach thus provides quantitative validation of Tomelleri's qualitative stemmatic analysis and offers a scalable methodology for CS textual criticism. By demonstrating that HTR-LLM-network workflows can reconstruct manuscript genealogies that align with traditional philology, this study expands the epistemological toolkit of paleoslavic studies and advocates for broader integration of CS sources into computational humanities research.

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Keywords: Handwritten Text Recognition, Large Language Models, Wenceslas of Bohemia, hagiography, Church Slavonic, data extraction